

ORR, MN, USA

WMO: 726544

Lat: 48.016N

Lon: 92.856W

Elev: 1311

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04958

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-18.5	-31.9	1.0	-24.6	-26.0	1.4	-18.0	19.5	9.9	17.8	9.8	3.1	300	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	84.5	69.4	81.9	67.4	79.5	65.6	73.5	80.8	71.0	78.7	68.7	76.0	8.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	123.1	79.2	68.4	109.7	75.5	65.9	100.4	73.1	38.0	80.1	35.6	79.1	33.8	77.1	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.1	13.5	-30.9	90.3	6.3	3.4	-35.4	92.7	-39.1	94.7	-42.6	96.6	-47.2	99.1		
(5)			-30.7	76.2	6.2	2.6	-35.1	78.1	-38.8	79.6	-42.3	81.1	-46.8	83.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	6.8	11.4	24.7	38.6	51.6	61.1	65.9	63.7	56.2	43.0	26.9	13.2		
	DBStd	23.20	14.88	14.06	13.42	10.17	8.78	7.04	5.88	6.10	8.74	9.10	11.46	13.41		
	HDD50	5786	1339	1080	786	361	84	5	0	1	37	258	694	1141		
	HDD65	9801	1804	1500	1248	792	424	156	60	99	286	683	1143	1606		
	CDD50	1677	0	0	3	18	134	337	494	426	222	42	1	0		
	CDD65	215	0	0	0	0	9	37	89	58	21	1	0	0		
	CDH74	2328	0	0	1	10	173	424	914	570	219	17	0	0		
	CDH80	540	0	0	0	2	40	94	231	125	45	3	0	0		
	WSAvg	4.9	5.1	5.0	5.2	5.1	5.3	4.5	4.2	4.1	4.7	5.5	5.6	4.8		
	PrecAvg	27.8	0.9	0.7	1.1	2.0	3.0	4.0	4.0	3.2	3.7	2.6	1.5	1.1		
(15) Precipitation	PrecMax	33.6	2.0	1.3	3.3	4.4	5.1	11.1	8.0	6.2	7.4	4.3	3.4	2.3		
	PrecMin	23.1	0.4	0.1	0.4	0.6	1.1	1.7	1.7	0.7	1.2	0.6	0.1	0.6		
	PrecStd	3.1	0.4	0.4	0.7	0.9	1.1	2.0	1.9	1.5	1.8	1.1	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	45.9	63.4	73.2	84.2	88.1	90.4	88.1	84.3	75.2	57.3	41.5		
		MCWB	34.9	37.6	50.2	55.1	62.1	72.2	73.6	71.3	68.5	59.7	49.2	36.4		
	2%	DB	34.5	39.4	54.2	66.4	79.2	81.9	84.8	82.5	80.7	68.2	52.4	36.8		
		MCWB	32.2	34.7	44.8	50.5	59.9	67.0	71.1	68.3	67.0	56.8	46.9	34.0		
	5%	DB	30.4	36.0	47.7	62.7	73.2	79.1	81.9	80.8	75.2	63.0	46.2	34.1		
		MCWB	29.3	32.9	40.5	47.4	56.3	64.8	68.5	67.0	64.6	54.0	41.6	32.1		
	10%	DB	27.5	31.9	43.0	56.7	69.7	75.0	79.4	76.9	71.9	56.9	42.9	30.3		
		MCWB	26.0	29.7	36.7	44.7	55.5	63.1	66.6	65.5	62.4	50.3	38.6	29.4		
	0.4%	WB	36.4	39.9	54.8	57.7	68.4	75.2	76.9	75.2	74.2	64.6	53.9	37.4		
		MCDB	39.5	44.3	58.5	68.9	76.7	82.8	85.4	82.7	82.3	68.2	55.7	39.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.5	36.1	46.0	52.9	63.5	71.0	74.0	71.5	69.9	59.1	46.3	34.3		
		MCDB	34.3	39.2	52.0	62.6	72.7	79.4	81.6	79.1	76.3	65.4	50.3	36.6		
	5%	WB	29.5	32.9	41.2	49.1	59.9	68.1	71.5	69.2	66.4	54.8	42.5	31.9		
		MCDB	30.7	35.6	47.3	59.9	69.8	74.9	79.3	77.5	72.6	62.1	46.5	33.7		
	10%	WB	26.2	29.2	37.2	45.6	56.9	65.6	68.9	67.1	63.5	51.5	38.6	29.6		
		MCDB	27.5	31.5	42.1	54.6	65.7	72.1	76.3	74.1	69.3	56.1	42.4	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	21.1	21.4	22.3	23.8	22.4	22.8	22.6	21.5	16.9	14.3	15.2		
		MCDBR	17.1	20.8	24.7	31.7	31.6	27.0	26.0	26.2	26.6	25.3	19.7	16.1		
		MCWBR	14.7	15.8	15.9	18.3	16.0	14.3	13.5	13.2	14.9	15.1	13.7	13.3		
		MCDBR	16.1	17.9	22.4	27.8	25.8	22.5	23.7	22.4	22.3	22.5	18.6	15.2		
	5% WB	MCWBR	14.1	13.9	15.3	17.7	14.8	13.9	14.4	12.8	14.4	14.9	13.9	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.266	0.315	0.341	0.357	0.372	0.387	0.372	0.356	0.306	0.282	0.260		
	taud		2.360	2.327	2.343	2.395	2.407	2.384	2.361	2.436	2.452	2.581	2.519	2.394		
	Ebn at Noon		266	287	287	289	287	282	276	276	270	271	255	248		
	Edh at Noon		22	28	33	35	36	37	37	33	30	22	19	18		
(44) All-Sky Solar Radiation	RadAvg		430	801	1182	1494	1666	1770	1887	1598	1182	694	391	304		
	RadStd		54	79	111	143	115	138	96	115	95	95	50	31		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.80	-2.74	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page